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Improving Integrity in Voluntary Carbon Markets through Technological Innovation

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Abstract

The voluntary carbon market (VCM) is projected to reach \$250 billion by 2050. However, the market faces a confidence crisis, with dramatic declines in transaction volume and pricing. Realizing this potential is related significantly to technological innovations addressing data quality, governance, and transparency challenges.

This research targets individuals and organizations developing technological solutions to enhance carbon market integrity. It provides an accessible introduction to carbon markets and evaluates how emerging technologies can address market failures through integrated systems.

The study examines efforts by integrity initiatives and private sector actors to address limitations through technological innovation, focusing on digital monitoring, reporting and verification (D-MRV), blockchain applications, artificial intelligence, and standardization approaches.

Semi-structured interviews were conducted with market participants across registries, rating agencies, and technology companies. Thematic analysis is presented alongside academic literature, company reports, and organization publications.

The research reveals that data integrity serves as the foundational prerequisite for all technological solutions, with consensus that digital monitoring must replace manual verification methods. While blockchain shows promise for transparency, energy consumption and fragmentation risks limit adoption. Large language models demonstrate applications in data aggregation and regulatory analysis, though technical barriers remain. Market convergence creates opportunities for government-centered solutions.

The study recommends prioritizing data integrity solutions, focusing on government technology innovations for sovereign oversight, and designing community-centered technologies creating local employment. Strategic focus should target engineered removal solutions and enhanced corporate disclosure through user-friendly platforms building market expertise.

Nomenclature

AI	Artificial Intelligence
CCP	Core Carbon Principles
CDM	Clean Development Mechanism
COP	Conference of Parties
CORSIA	Carbon Offsetting and Reduction Scheme for International Aviation
D-MRV	Digital monitoring, reporting and verification
DBH	Diameter at breast height
DLT	Distributed ledger technology
DVP	Delivery versus payment
ESG	Environmental, Social, and Governance
ET	Emissions Trading
GIS	Geographic Information Systems
ICVCM	The Integrity Council for the Voluntary Carbon Market
IoT	Internet of Things
IPCC	Intergovernmental Panel on Climate Change
ITMO	Internationally Transferred Mitigation Outcomes
JI	Joint Implementation
KDA	Kinexys Digital Assets
LiDAR	Light Detection and Ranging
LLM	Large Language Models
NDC	Nationally Determined Contributions
OECD	Organisation for Economic Co-operation and Development

OTC	Over-the-counter
PDD	Project Design Document
REDD+	Reducing emissions from deforestation and forest degradation
RFP	Request for Proposal
SBTi	Science Based Targets initiative
TSVCM	Taskforce for Scaling the Voluntary Carbon Market
UNFCCC	United Nations Framework Convention on Climate Change
VCC	Voluntary carbon credits
VCM	Voluntary carbon markets
VCMI	Voluntary Carbon Markets Integrity Initiative
VCS	Verified Carbon Standard
VVB	Validation and Verification Bodies

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1 Introduction

(Section 1.1 is written as a general primer, situating this research within the wider climate narrative. Readers already familiar with these foundations may wish to proceed directly to Section 1.2.)

1.1 A Brief History of Climate Change and Climate Finance

Climate change has been one of the most pressing environmental and societal risks of the 21st century, indicated by global temperature increase caused by excessive greenhouse gas emission to the atmosphere [1]. This is supported by the **Intergovernmental Panel on Climate Change (IPCC)**, an international body which has provided regular scientific assessments on climate change since 1988, demanding urgent, coordinated international cooperation across borders and sectors.

The **United Nations Framework Convention on Climate Change (UNFCCC)** is the first international framework adapted in 1992 under which ratifying countries, dubbed "Parties to the Conventions", convene annually at the **Conference of Parties (COP)**, to discuss strategies in preventing "dangerous" human interference with the climate system by stabilizing greenhouse gas emissions [2]. The UNFCCC was operationalized by the **Kyoto Protocol** in 1997, which binded developed countries to adopt policies and measures on mitigation and to report periodically. Negotiations under the UNFCCC stagnated for over twenty years, due to disputes concerning historical injustice and responsibilities over global warming [3]. This stalemate persisted until the adoption of the **Paris Agreement** in 2015, the first legally binding international agreement under the UNFCCC applicable to all Parties, with the overarching goal of "holding the increase in the global average temperature to well below 2 °C above pre-industrial levels and pursuing efforts to limit the temperature increase to 1.5 °C above pre-industrial levels" [4], through the setting of **Nationally Determined Contributions (NDCs)**, which are emission targets set by countries themselves, to be submitted to the UNFCCC secretariat once every 5 years from 2020 onwards.

Addressing the climate crisis via NDCs requires not only political will but also a fundamental shift in capital allocation, giving rise to the field of **climate finance**, which refers to the mobilization of public and private financial flows specifically aimed at supporting climate mitigation and adaptation goals [5]. Alongside mechanisms such as dedicated climate funds, green bonds, and blended finance, carbon markets have emerged as a promising approach with significant potential to enhance the effectiveness and reach of climate finance.

1.2 Basics of Carbon Markets and Carbon Credits

Carbon markets are all about generating a carbon price as signal to simulate economically efficient decarbonization, via the trading of carbon credits, tradable certificates that each represents the removal or reduction of one metric ton of greenhouse gas emissions from the atmosphere. These credits can be bought and sold in carbon markets, enabling companies and individuals to compensate for their own emissions while channeling finance into projects that deliver measurable climate benefits, beyond its own value chain.

Carbon credits can be classified by the type of mitigation. Some credits result from emission reductions and avoidance, reducing greenhouse gases released or preventing their release from production altogether, while others represent removals, where carbon dioxide is extracted from the atmosphere and durably stored. The distinction is important for both project developers and buyers seeking specific climate outcomes.

Both types of mitigations can be achieved through either nature-based or technology-based solutions. Nature-based projects include afforestation and reforestation (removal), avoided deforestation (reduction), soil carbon enrichment from improved farming practices (removal), and the conservation of blue carbon ecosystems like mangroves and salt marshes (removal). Technology-based solutions include renewable energy generation such as wind or solar (reduction), methane capture from landfills or agriculture (reduction), and engineered removals like direct air carbon capture and storage (removal) [6].

1.3 Importance of Carbon Price

Pricing carbon means internalizing the social and environmental costs of greenhouse gas emissions by attaching a monetary value to each ton released. Carbon pricing is central to climate policy because rising carbon prices create stronger incentives to cut emissions and shift capital toward low-carbon pathways [7, 8]. The recommended carbon price that addresses environmental and social costs for 2025 is approximately \$100–\$200 per ton of CO₂e, according to expert consensus in recent global economic and scientific literature [9]. Yet current prices remain far below levels aligned with climate goals [8]. Price determination also carries risks: volatility, thin demand, or weak governance can lead to distorted signals and undermine trust [10]. For credibility, the price must further reflect the quality of emissions reductions, ensuring that high-integrity projects command higher values. Without such differentiation, prices risk rewarding low-impact credits. This makes it vital to strengthen how carbon price

dynamics balance level, stability, and integrity to guide effective climate finance.

1.4 Distinction between the Compliance and Voluntary Market

The compliance market primarily operates under government-mandated cap-and-trade schemes, in which companies must hold and trade carbon allowances, with each allowance representing the right to emit one tonne of CO₂.

The voluntary carbon market (VCM), by contrast, is driven by organizations pursuing self-imposed climate commitments, and is not mandated by law. Instead, it is guided by private standards such as Verra's Verified Carbon Standard (VCS) or the Gold Standard, which set the rules for crediting, issuance, and retirement.

A source of confusion arises because the boundary between the two markets is becoming increasingly blurred. Some compliance systems now allow the use of carbon credits to meet a share of mandatory reduction obligations (for example, under CORSIA or Article 6 of the Paris Agreement). In these cases, credits that were originally designed for voluntary offsetting are effectively "imported" into a legal compliance framework, where they are allocated, traded, and retired under stricter regulatory oversight. This overlap makes it difficult for stakeholders to distinguish between credits that carry purely voluntary value and those that have compliance relevance.

In recent years, there has been rapid proliferation of corporate net-zero and carbon neutrality targets, largely fueling the growth of the voluntary carbon market. Initiatives like the Science Based Targets initiative (SBTi) urge companies to set robust climate goals and, where emissions cannot be immediately eliminated, to use voluntary credits to credibly offset their residual emissions, aligning their corporate strategy with a 1.5°C future. Nearly every major company and a growing number of institutions now claim climate neutrality or net-zero ambitions.

1.5 Voluntary Carbon Market (VCM)

1.5.1 A brief history of the VCM

Although the Voluntary Carbon Market (VCM) has only gained prominence in the past decade, its history dates back to the late 1990s, predating the emergence of compliance markets and the adoption of the Kyoto Protocol. The Kyoto Protocol introduced three separate market-based mechanisms to incentivize environmentally friendly behavior. These include the Clean Development Mechanism (CDM), where developed countries can buy credits from projects in developing countries; the Joint

Implementation (JI), allowing developed countries to earn credits from projects in other developed countries; and Emissions Trading (ET), which has now become a cornerstone of carbon markets.

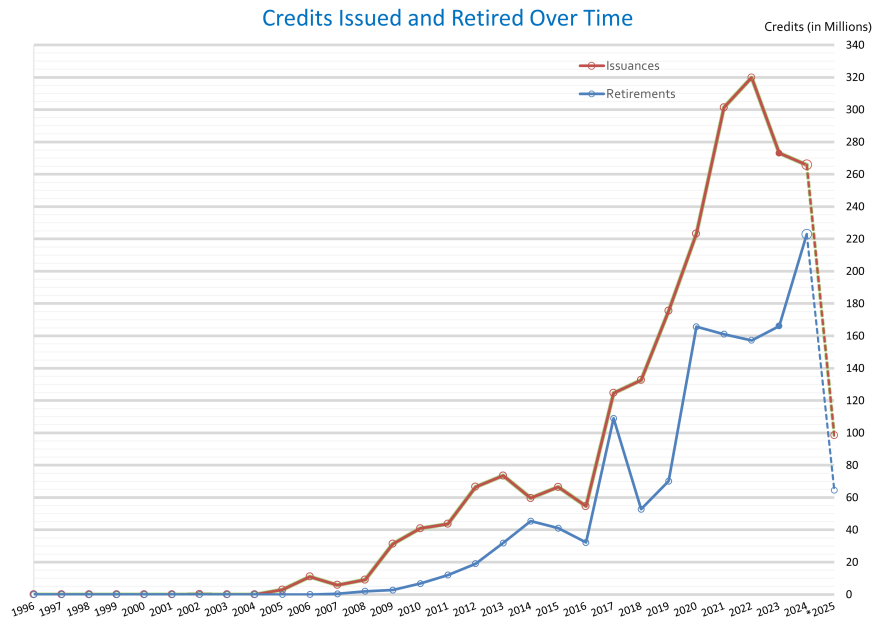


Figure 1: Credits issued and retired over time as of April 30, 2025 [11]

From 2007 to 2016, the demand for carbon offsets grew partly due to high-profile commitments by major corporations such as Microsoft and The Walt Disney Company to achieve carbon neutrality. However, it was only following the Paris Agreement in 2016 and the introduction of the first Corporate Net-Zero Targets by SBTi in 2021 that the voluntary market saw considerable growth, exceeding 100 million credits issued for the first time in 2017 and surpassing 300 million credits by 2022, illustrating strong momentum in the market, as seen in figure 1.

1.5.2 Key Actors in the VCM Credit Lifecycle

The carbon credit lifecycle involve 3 stages: **credit creation**, **intermediation** and **retirement**. Each stage involve multiple stakeholders, some of which may participate across more than one stage, as illustrated in figure 2. To illustrate this process clearly, we will follow the real-life lifecycle of a carbon offsetting project from inception to credit retirement.

On the supply side, **project owners**, **developers**, and **financers** initiate carbon projects aimed at reducing or removing atmospheric greenhouse gases. For example, in the *Katingan Mentaya Project (ID 1477)*, the project developer is *PT Rimba*

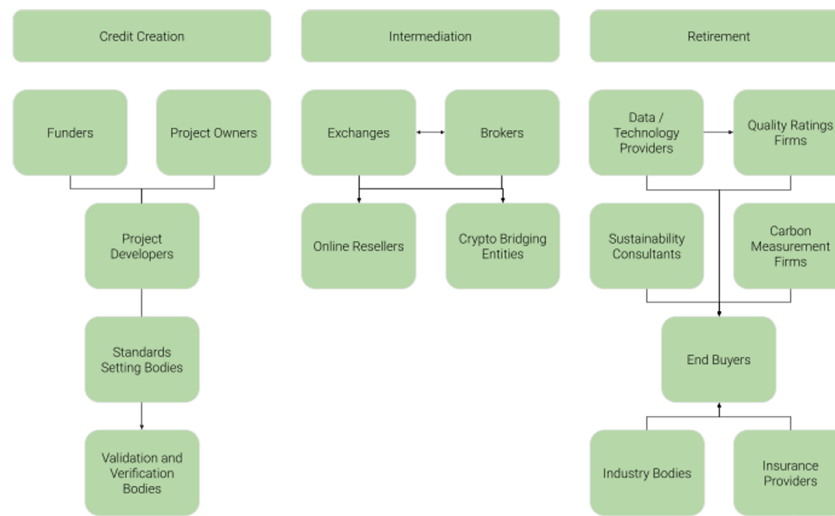


Figure 2: Credit lifecycle and stakeholders involved, adapted from Allied Offsets, 2023 [12]

Makmur Utama, which designed the project to protect peatland forest ecosystems in Central Kalimantan, Indonesia.

Project owners, developers and financiers proceed by selecting a **carbon registry** under which the projects would be registered. In the case of the Katingan Mentaya Project, *Verra's Verified Carbon Standard (VCS)* is chosen as the **carbon crediting program** under which their project would be registered. Verra, like other registries (e.g. Gold Standard, ACR), serves as a platform that track the issuance, trade, and retirement of carbon credits while providing a set of project-specific methodologies to calculate emissions reductions and ensure transparency throughout the project's lifecycle. Since the Katingan Mentaya project is a **nature-based solution**, it adopted a relevant "reducing emissions from deforestation and forest degradation in developing countries" (REDD+) methodology i.e. *VM0007* [13]. These methodologies are program-specific rules and are influenced by external **standard setting bodies** (e.g. *ICVCM*) and rating agencies (e.g. *BeZero Carbon*).

Project developers document their plans in a **Project Design Document (PDD)**, in accordance with the methodology requirements, detailing the project activities, baselines, and expected carbon benefits. The PDD is submitted to a **Validation and Verification Body (VVBs)**, who assess the project for eligibility, accuracy of baselines, and estimated emission reductions over time [14]. In the case of the Katingan Mentaya project, *SCS Global Services* were employed to independently assess the PDD against Verra's criteria. Once validated, the PDD is then submitted to the registry for a final review and approval before the project can be formally listed and

proceed to implementation.

Once reviewed and approved, projects are given the green flag to operate. During the operation process, project developers intermittently, usually annually, collect monitoring data, project reports and other documentations and submit them to a third-party verifier, who assesses the data and ensures that the projects meets all the requirements of the chosen carbon credit standard. If satisfied, the verifier issues carbon credits according to the amount documented, which can be traded. This cycle is repeated until the end of the project's operation. For the Katingan Mentaya project, *Aster Global Environmental Solutions* and *AENOR CONFIA, S.A. (UNIPERSONAL)* were subsequently employed for verification.

The generated carbon credits are primarily traded over-the-counter (OTC), facilitated by **brokers** and **advisory service providers**, but they could also be traded on exchanges (e.g. *Carbonplace, Xpansiv*). Notably, secondary market trade volumes, largely facilitated by online resellers and crypto-bridging entities, are estimated to exceed reported spot volumes by several multiples [15].

On the demand side, **end buyers** are motivated to purchase carbon credits as part of their efforts to meet climate goals, regulatory obligations, or broader ESG strategies. Buyers often select projects that align with their sustainability values by reviewing data from carbon credit rating agencies (e.g., BeZero, Sylvera) or directly accessing information from carbon registries. Alternatively, they attempt to fulfill their ESG initiatives by outsourcing **sustainability consultants** to engage with the VCM. To protect against project setbacks such as underdeliveries, **insurance companies** (e.g. Kita, Artio) provide risk-transfer instruments that guarantee compensation or replacement credits in the event of project underperformance, thereby de-risking investments and enhancing confidence in the market.

1.5.3 The Increasingly Blurry Line between Voluntary and Compliance Market

The boundary between voluntary and compliance carbon markets is increasingly blurred, driven by evolving international frameworks and rising climate ambitions. The Paris Agreement marked a paradigm shift from the Kyoto Protocol by establishing that all countries share responsibility for reducing emissions, particularly in their economically intensive sectors. Nationally Determined Contributions (NDCs) are country-level climate pledges, while not legally binding compliance obligations, represent country-wide commitments to reduce emissions. This nuance has contributed to the growing overlap between voluntary actions and compliance requirements.

Article 6 of the Paris Agreement, plays a pivotal role in this convergence. It intro-

duces a global mechanism for cooperative carbon market approaches governed by the United Nations, aiming to facilitate the trading of internationally traded credits, coined internationally traded mitigation outcomes (ITMOs). These ITMOs can potentially be used in sovereign compliance schemes, thereby linking voluntary market credits with compliance-like use cases and opening new pathways for carbon credit utilization beyond traditional markets. However, the full operationalization of Article 6 remains pending, hindered by concerns over carbon credit credibility and market integrity. Research such as the Taskforce for Scaling the Voluntary Carbon Market (TSVCM) report of 2021 highlights that credit integrity remains the primary concern among buyers, with 45% identifying it as a significant barrier to market engagement. The growing intersection of voluntary and compliance markets therefore brings both opportunity and the urgent need for enhanced transparency, robust verification, and harmonized standards to maintain trust in carbon markets as they evolve.

1.6 Knowledge Gap

Current studies on voluntary carbon markets (VCMs) have primarily concentrated on themes such as policy frameworks, governance, and the role of carbon pricing in achieving climate goals. For instance, the Organisation for Economic Co-operation and Development (OECD) has recently published high-level guidance exploring the role of governments in strengthening credit integrity [16]. Numerous studies have also analyzed the performance of projects under mechanisms like the Clean Development Mechanism (CDM), emphasizing issues of additionality, permanence, and co-benefits. In recent years, research has increasingly turned toward the geopolitical and institutional dimensions of VCMs, including the interaction with Article 6 of the Paris Agreement and the integrity of offset claims.

However, bibliometric evidence suggests that while there is extensive discussion on policy and governance aspects, there remains a significant gap in examining how technological innovation can be systematically leveraged to improve the integrity of carbon credits [17]. Specifically, limited progress has been made on synthesizing approaches from fields such as natural language processing, blockchain, or multi-agent systems into the assessment and monitoring of carbon credit quality, while co-existing with existing infrastructure. What is missing, therefore, is a comprehensive investigation that bridges traditional integrity debates with emerging technological solutions, by synthesizing insights from diverse industry actors to avoid fragmentation.

Addressing the gap as part of continuous improvement is crucial [18], because without a clear understanding of these implications, key stakeholders may struggle to make informed decisions about adopting or supporting such technologies, leading to missed opportunities, slow adoption rates, or worse, unintended adverse effects on credit pricing and liquidity.

1.7 Research Outline & Questions

Building on this, the present research focuses on how technological innovation can strengthen integrity in voluntary carbon markets, aiming to provide students and industry professionals with the knowledge needed to develop tools that enhance these markets. In short, there is a need for better understanding of the state, limitation, and opportunities in carbon credit integrity.

Accordingly, this research seeks to address the following questions:

1. What are the key integrity failures undermining voluntary carbon markets today? (**RQ1**)
2. How can emerging technologies address these specific market failures, and work together as an integrated system? (**RQ2**)
3. What are the implementation challenges and scaling requirements for deploying these technological solutions effectively? (**RQ3**)

2 Integrity Concepts & Challenges

2.1 Integrity in Credit Creation (Supply Side)

Supply-side integrity is all about the quality of the credits, more precisely the ability of the underlying project to deliver its intended climate benefit. It is best explained by the **Core Carbon Principles (CCPs)** established by the ICVCM, which is a framework endorsed by more than 250 international organizations. Launched in March 2023, the CCPs establish ten fundamental criteria organized into three categories: Governance, Emissions Impact, and Sustainable Development. These principles create a global benchmark for high-integrity carbon credits [18]. Supply-side integrity can be simplified down to 3 criteria: additionality, permanence, and carbon accounting.

Additionality serves as perhaps the most critical criterion in supply-side integrity, requiring that emission reductions or removals would not have occurred without the fi-

nancial incentive created by carbon credit revenues. For instance, a large-scale solar farm in a region where renewable power is already the cheapest option, supported by government subsidies and strong electricity demand, would lack additionality, as it would proceed without carbon credit revenue. By contrast, a project that deploys solar microgrid in a remote village, where high infrastructure costs and limited demand make the project financially unviable, demonstrates clear additionality, since carbon finance is essential for its implementation [19]. Carbon crediting programs employ various **additionality tests** to assess projects, including financial analysis (demonstrating that projects are not viable without carbon revenue), and common practice analysis (ensuring projects are not standard industry practice), which are then verified by third-parties [20].

Permanence addresses whether emission reductions or carbon removals are sustained over time without being reversed. This concern is especially relevant for nature-based solutions. For example, a reforestation project may store carbon in trees, but that carbon could be released back into the atmosphere if the forest burns in a wildfire or is later cleared for agriculture. To manage this risk, most carbon crediting programs use **buffer reserves** or **risk pooling mechanisms**. A portion of credits, for example, 5% [21], is withheld from sale and placed in a shared reserve that functions like insurance. If a reversal occurs, credits are cancelled from this reserve to ensure that buyers' claims remain valid. In effect, projects can only sell the verified reductions or removals minus the buffer contribution and other deductions, which protects the integrity of the system [22].

Carbon accounting is the process of quantifying and verifying emissions reductions or removals using transparent and rigorous methodologies. It ensures that carbon credits represent genuine climate benefits by accurately modelling both the baseline scenario, what emissions would have been without the project, and the project scenario, which tracks actual emission reductions achieved over time, drawing on historical data and regional context. An important part of carbon accounting also involves assessing leakage, unexpected losses of anticipated carbon benefits while a project is being executed, where emissions may be displaced outside the project boundaries. For example, in avoided deforestation projects (REDD+), historical deforestation rates establish the baseline, while ongoing monitoring of forest cover measures the project's impact. Leakage assessment looks at whether deforestation is simply shifted to surrounding areas rather than reduced overall.

2.2 Integrity in Credit Retirement (Demand Side)

While much attention in voluntary carbon markets (VCM) has focused on improving supply-side integrity through enhanced methodologies and verification, the demand side remains comparatively less discussed but is equally crucial for market legitimacy. Demand-side integrity addresses how buyers use and retire carbon credits, ensuring that claims about emissions reductions are credible and not double counted, meaning each credit is used only once to support a climate claim. Several important frameworks guide credible demand-side credit use and transparency. Most prominently, the **Voluntary Carbon Markets Integrity Initiative (VCMI) Claims Code** provides standards for using credits in climate plans aligned with scientific decarbonization targets [23]. Alternatively, the **Oxford Offsetting Principles** proposed ethical and practical guidelines for offsetting [24].

Specialized insurers and consultants play critical roles in demand-side integrity by assessing risks, validating the legitimacy of carbon claims, and advising entities on credit selection, retirement timing, and accounting methodologies. They help ensure buyer confidence and facilitate adherence to evolving best practices [25]

Challenges on the demand side include uneven buyer understanding, fragmented retirement protocols across standards, and limited real-time transparency of retirements. There is risk of overlapping assessments by rating agencies, insurers, and consultants, which, while providing safeguards, can also create inefficiencies. Advancing technology solutions and targeted capacity building can help unify demand-side integrity efforts, improving market efficiency and trust overall.

3 Methodology

The VCM's inherent characteristics, rapid evolution, data fragmentation, opacity and predominant OTC trading, created fundamental challenges for quantitative analysis. This necessitated a qualitative research approach, with semi-structured interviews chosen.

Semi-structured interviews use an interview guide to outline the key themes relevant to the study, while allowing flexibility for participants to elaborate on their experiences and perspectives. This balance between structure and openness is particularly suited for exploring emerging fields such as the VCM, where expert insights can provide context, uncover nuanced challenges, and focus on the issues that were meaningful for the participant. To ensure methodological rigour, the study follows a 3-step framework for developing semi-structured interview guides adapted from Kallio et al. [26].

Table 1: Three-phase framework for developing semi-structured interview guides (adapted from [26])

Phase	Description
Phase 1: Identifying prerequisites	Assessing the appropriateness of semi-structured interviews for the research. Suitable for studying participants' perceptions and opinions on complex issues, with flexibility to focus on themes meaningful to the participant.
Phase 2: Retrieving and using previous knowledge	Incorporating insights from prior research and related frameworks.
Phase 3: Formulating the preliminary interview guide	Operationalizing key themes into specific questions. The interview guide lists main themes and follow-up questions, allowing flexible order during the interview while directing the discussion to research-relevant areas. Refer to Appendix B for the interview guide.

The qualitative data analysis followed a structured, iterative process. Interview transcripts were manually coded using Atlas.ti [27]. The coding strategy was inductive and deductive, designed to identify segments of text relevant to the research questions. These codes were subsequently synthesized through thematic analysis to identify recurring patterns and central themes across the dataset.

The findings will be presented as a **thematic narrative**. To ensure analytical rigor and trustworthiness, the analysis will be supported by direct illustrative quotations from the interviews. These primary data will be triangulated with evidence from the

existing academic literature, industry publications, and relevant intergovernmental reports to contextualize and validate the emerging themes.

4 Results and Discussion

(The complete list of interviewees are tabulated in Appendix A.)

4.1 General Market Outlook

4.1.1 Market Performance Overview

The voluntary carbon market has significantly underperformed against earlier growth projections. While McKinsey and TSVCM predicted the VCM would scale 15-fold to \$30 billion by 2030, assuming 1-2 gigatonnes of CO₂ demand [28], current market reality presents a more constrained picture.

The market experienced its lowest transaction volume since 2018 in 2024, with total carbon credit transaction volume declining by 25% compared to 2023 and market value dropping to \$535 million [29]. This continues a contraction trend, with 2023 showing a 56% decline in overall market transaction volume [30].

However, despite lower volumes, market value remains 1.9 times higher than 2018 levels due to relatively buoyant credit prices. This price resilience suggests a fundamental shift toward quality over quantity, with buyers increasingly selective about credit purchases amid ongoing integrity concerns and market maturation.

4.1.2 Market Credibility

All participants agreed that the VCM suffers from credibility challenges that undermine its effectiveness as a climate finance mechanism. Multiple respondents mentioned that the global voluntary carbon markets have been thrown into crisis by fraud and quality scandals over the past few years. These scandals span across project types, including, the Kariba REDD+ Project's rating collapse from AA+ to removal from coverage [31], and 90% of Verra's rainforest credits failing to represent real reductions [32], and CQC's cookstove project that led to an overissuance of 5 million VCU's [33].

As the most frequently identified factor influencing credit demand from sentiment analysis [29], media coverage plays a crucial role in holding the carbon market accountable by exposing fraudulent practices, but this scrutiny has amplified market concerns over greenwashing accusations, making projects hard to gain traction.

A critical weakness in market integrity stems from insufficient organizational capacity, as Simon Puleston Jones, CEO of Emral Carbon, explained, *"85% of carbon credit*

buyers do not have a person within that company that is responsible for their engagement with the carbon market.” This staffing gap reflects broader under-investment in carbon market expertise.

4.1.3 Market Governance

The credibility crisis extends beyond individual project failures to systemic governance breakdowns, particularly in compliance with international climate commitments. As Jaye Connolly, CEO of RippleNami, pointed out, “[...] *only 13 out of the 195 countries that ratified the Paris Agreement managed to meet the February 2025 deadline for submitting updated NDCs.*” This governance gap creates a disconnect between international climate ambitions and the institutional frameworks, which trickles down to affect the overall credibility sentiment towards the VCM.

The market credibility crisis represents more than isolated scandals, but systematic failures in governance, capacity, and oversight that threaten the carbon market’s role in scaling climate finance. While these spectrums cannot directly be addressed through technological advancements, understanding the context is vital for feasibility assessments of technological solutions.

4.2 Emerging technologies

4.2.1 Data

All respondents agreed that emerging technology would play an important role in giving confidence in data, and demonstrated consensus that highlights data as the fundamental building block upon which all other technological solutions depend.

Jaye pointed out how there are many data silos in the VCM, with fragmented due diligence often resulting from proprietary methodologies. The current state of the VCM is heavily consultant-driven, which has resulted in huge piles of paperwork and an overflow of unstructured data that limits scalability and transparency.

All respondents reported support for Digital Monitoring, Reporting, and Verification (D-MRV) approaches to replace archaic random sampling methodologies, such as field surveyors manually measuring tree diameter at breast height (DBH) on randomly selected plots and recording data in spreadsheets. This technological shift represents a fundamental change from periodic, manual verification to continuous, automated data collection and analysis.

Simon emphasized that data confidence enables innovation in the provision of insurance solutions, demonstrating how improved data quality creates downstream

opportunities for financial product development and risk management in carbon markets.

4.2.2 Blockchain

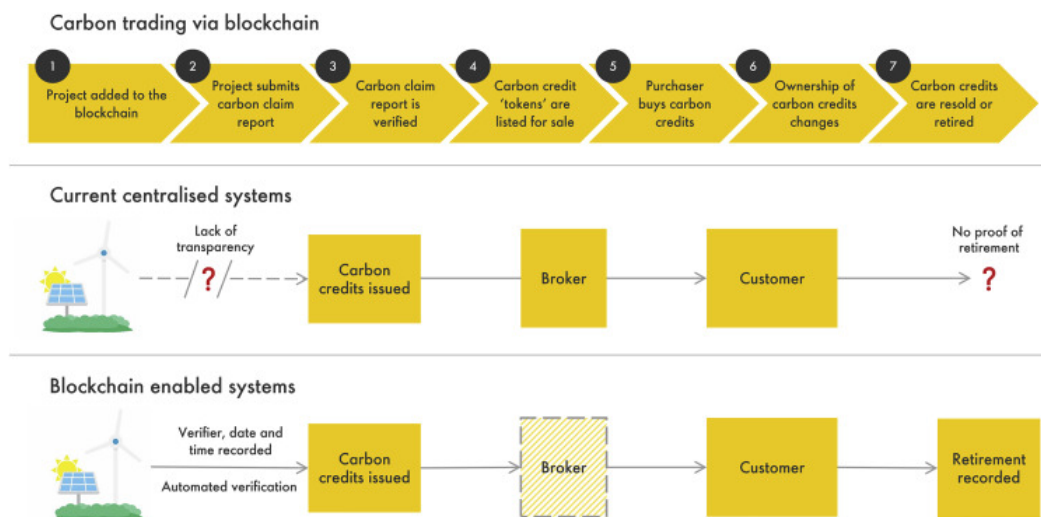


Figure 3: Schematic overview comparing current centralized systems and blockchain-enabled systems, reproduced from [34].

Blockchain-based solutions are increasingly being proposed by companies and organizations in carbon markets. These solutions leverage immutable transaction records, eliminate intermediaries through smart contracts, and enable transparent proof of credit retirements and transfers, as illustrated in Figure 3. Companies, such as Moss.earth, KlimaDAO, have adopted blockchain-based solutions to drive the transition as early as since 2021 [35]. Among recent propositions, Kinexys Digital Assets (KDA), a multi-asset tokenization platform from JPMorgan Chase, began native tokenization for carbon credits since July 2025 [36]. Notably, september 2025 saw Zimbabwe becoming the first country to launch a national carbon registry using blockchain technology [37]. These developments illustrated both jurisdiction-level adoption and growing optimism from major financial institutions.

The market now features several fully competitive blockchain solutions. The earliest competitive example is the Toucan Protocol, which takes credits from the Verified Carbon Standard (VCS), retires them, and mints a replica token on the blockchain known as a Base Carbon Tonne that can be transferred or sold as required [38]. However, Simon noted that most registries don't use blockchain throughout the life of the project, with only some projects utilizing distributed ledger technology (DLT) end-to-end. Even when registries claim to use DLT and credits are issued as digital assets, the underlying project data is often not stored on a distributed ledger.

Blockchain enables delivery versus payment (DVP), which eliminates principal risks while reducing the need for collateral enabled by stablecoins. This is particularly relevant given regulatory efforts shifting to grant some legitimacy to blockchain, such as the US GENIUS Act signed into law in July 2025, which created the first federal regulatory framework for payment stablecoins by requiring 100% reserve backing and establishing clear oversight standards [39].

Despite this proliferation, multiple respondents demonstrated skepticism about applying blockchain in carbon markets, though for different reasons.

The energy cost of blockchain solutions emerged as a significant counterargument to the blockchain narrative. Jaye noted that *“blockchain is expensive and kludgy.”* While DLT platform providers in 2025 often emphasize commitments to carbon neutrality, with examples including Algorand, Hedera, Cardano, and Polkadot using low-energy consensus mechanisms and carbon offsetting [40], as Scott Eaton, CEO of Carbonplace, explained, blockchain use is not environmentally sound, stating that blockchain is *“an incredible energy consumer. So to use it in a carbon context doesn’t really give a consistent message.”* This messaging inconsistency, reinforced by the paradox of using carbon offsets to justify blockchain’s environmental impact in carbon markets, is one reason why some organizations including Carbonplace has moved away from blockchain and re-skinned on more traditional tech stacks, despite having spun out with the intention of leveraging blockchain.

Just as there are concerns over too many standards, respondents expressed fears over blockchain fragmentation. Scott noted the need for general adoption within financial services and agreement on which blockchain to use, because *“the bridge is the easiest to hack point”* and creates vulnerabilities for double counting.

This skepticism is supported by market evidence, including Verra’s May, 2022 announcement to pause on-chain credit retirements and attract public consultation due to confusion amongst buyers and double-counting risks [41]. To date, Verra is still working on immobilization tokenization but has stated they will only move forward once confident about governance, fraud prevention, and double-counting safeguards.

Ultimately, the market for blockchain remains immature and still developing, with most DLT solutions demonstrating low technology readiness levels. Plus, it is generally agreed that end-to-end verified high quality data is the prerequisite for blockchain adoption in carbon markets.

4.2.3 Artificial Intelligence (AI)

All respondents reported use of Large Language Models (LLMs), most notably Perplexity AI, within their organizations for solving information blindspots and accessing market statistics. The technology has found practical applications across multiple carbon market functions, from sourcing to regulatory analysis.

Simon described using LLMs for *“conducting a RFP (Request for Proposal) on their (carbon credit buyers’) behalf to go source carbon projects that meet the criteria that they have set”*, demonstrating how AI streamlines the project identification and procurement process.

AI for data aggregation has also emerged as a key use case, with companies like BeZero Carbon leading practical applications. Joel Gould, market and policy lead at BeZero Carbon, explained that BeZero has been investing heavily in AI capabilities that *“help us really better aggregate and use data from projects in a more effective way and use that to interrogate the types of risk analytics that we provide.”* The company has successfully applied LLMs for sector classification and entity extraction tasks, demonstrating how AI can structure the wealth of unstructured project documentation in the VCM [42]. However, Joel noted that success depends on data quality: *“historically, an issue with some projects has been data collection has been pretty shaky, hasn’t been recorded very well, hasn’t been recorded in a standardized way”*, requiring massive teams of data engineers to *“bring inputs together and putting it in a sensible format that our research team can actually use.”*

On a governance level, Simon explained how LLMs assist *“in helping governments that are a party to the Paris Agreement in getting the capacity they need quickly and getting the product they need, so draft regulation and so on, at speed, so they can then participate in this market, which could give those countries, typically in emerging markets, access to billions of dollars each year thanks to that carbon market.”* Simon noted that the primary value lies not in education but in filtering complex information streams. As Simon noted, decision-makers are *“overwhelmed with opportunities to deploy their capital. What’s hard for them is making the right decisions based on their own governance and investment criteria and spotting issues, spotting risk, as they choose which projects to deploy capital into.”* This filtering capability enables governments to establish rules and regulations that foster domestic market growth.

Looking toward future applications, multiple respondents identified significant potential for AI in compiling project design documents (PDDs) as well as for automated verification of project compliance with methodologies by processing PDDs and raw IoT sensor data against established standards. This represents a fundamental shift

from manual compliance checking to automated verification, potentially addressing one of the most time-consuming and error-prone aspects of carbon project validation and creating continuous monitoring systems that could significantly enhance market integrity.

However, several technical barriers remain that must be addressed for these applications to reach full potential. Key limitations include limited context windows that restrict the amount of project documentation that can be processed simultaneously, hallucinations where LLMs fabricate information, and failure to handle tabular data and multi-modal data such as GIS information [42]. These constraints suggest that while LLMs show promise for structuring unstructured text data, successful deployment will require hybrid approaches that combine AI capabilities with traditional data processing methods and human oversight.

4.2.4 Technology-based credits

Technology-based credits constitute approximately 60% of all generated carbon credits [6], predominantly comprising renewable energy and fuel switching projects, followed by waste disposal and chemical engineering carbon capture initiatives [29]. However, the integrity and effectiveness of these solutions vary significantly across different project types.

Clean cookstove projects, an example of an energy efficiency project, distribute efficient stoves to households in developing regions to reduce fuel consumption. However, research has demonstrated significant integrity concerns, with studies indicating that cookstove projects overstate their climate benefits by 1,000% on average [43]. Jaye's fieldwork corroborated these findings, revealing fundamental usability issues when speaking to end-users in affected regions. As she explained, *"housewives reflected inability to fit their existing cookery on cookstoves, nor does it serve the purpose of filling an entire family (of 7)."* This disconnect highlights a critical failure to consider grassroots-level usability and actual adoption rates, demonstrating how technological solutions can fail without proper community consultation and design adaptation.

Renewable energy credit prices remain consistently low due to declining technology costs and widespread market adoption [29]. This indirectly created significant additionality concerns. These projects, predominantly concentrated in Asia, often would proceed regardless of carbon finance due to favorable economics, undermining their environmental integrity. The low pricing reflects market recognition of limited additional climate benefits beyond business-as-usual development.

Direct air capture (DAC) and **bioenergy with carbon capture and storage (BECCS)** offer more measurable and permanent results compared to traditional offset projects. These technologies attract investment from high-margin companies, particularly in the technology sector, willing to pay premium prices for high-certainty carbon removals.

Current costs remain prohibitive for widespread adoption. DAC costs range from \$400-1000 per tonne CO₂, with current operational facilities like Climeworks in Iceland operating at \$1000-1300 per tonne [44]. BECCS presents more favorable economics, with expert assessments projecting costs of \$62-200 per tonne CO₂ sequestered [45].

However, corporate buyers view these investments as future-proofing against regulatory scrutiny, as governments and regulators become increasingly suspicious of low-integrity credits such as renewable energy projects with questionable additionality as mentioned previously. The appeal lies in permanence and verifiability. Unlike nature-based solutions vulnerable to reversal through fire or deforestation, engineered systems provide geological storage with permanence measured in centuries to millennia [46]. This certainty justifies premium pricing for companies seeking credible net-zero claims and compliance with evolving regulatory frameworks that may favor permanent removals over temporary sequestration.

Biochar represents a particularly promising technology-based solution that aligns with just transition principles. Created through pyrolysis of biomass at elevated temperatures, biochar is a porous, high-carbon form of charcoal that simultaneously sequesters carbon while providing agricultural co-benefits including improved soil water retention and nutrient aggregation [47].

Permanence studies, including long-term soil incubation experiments, demonstrate that approximately 70-80% of biochar carbon remains recalcitrant after 100 years, offering significantly higher permanence compared to other soil organic carbon forms [48]. This durability addresses one of the core integrity challenges facing nature-based solutions.

The technology aligns with just transition principles through multiple co-benefits: improved soil quality enables domestic farmers to diversify income streams through enhanced agricultural productivity, while the production process can utilize agricultural waste that might otherwise be burned or discarded. This creates local employment and business (fertilizer) opportunities and addresses waste management challenges simultaneously.

However, implementation challenges remain significant. Access to suitable pyrolysis equipment is limited, with constraints on biomass dimensions and fractions that existing boilers can handle. Operational requirements include dedicated personnel, and technical issues can cause production disruptions if equipment is handled incorrectly. Additionally, opportunities exist to capture excess heat through thermal recovery systems, which could improve overall project economics and energy efficiency [47].

Despite these operational challenges, biochar demonstrates significant potential for improvement through technological innovation, including models adapted to different biomass types, pyrolysis conditions, and regional soil characteristics, particularly for smaller-scale projects and emerging producers [49].

4.2.5 D-MRV

Most participants mentioned the proliferating D-MRV solutions and data aggregation strategies, which is one that Joel emphasized as part of the value proposition that BeZero, amongst other rating agencies, provides. Remote sensing data had been made possible facilitated by collaboration with different organizations. Space agencies provides medium to coarse resolution data over long historical durations, while commercial partners enable higher resolution imagery, which is useful for monitoring canopy density, planting, and degradation, when coupled with deep learning. Closer to surface, airborne LiDAR surveys could unveil the structure of vegetation, from which canopy height, cover and biomass accounts could be achieved. In-situ ground measurements validate image-based estimations and provide uncertainty data for measurements. The data stack is illustrated in figure 4.

4.3 Standardization

Over the past few years, market actors have worked to develop the new project types, credit standards, governing boards, and independent quality evaluators that the VCM will need to sustain and scale for the next decades of climate action [29]. However, this explosion of new tools attracted mixed feelings amongst respondents, particularly relevant to the discussion regarding standardization.

4.3.1 Taxonomy Standardization

Following its consultation on sustainable finance frameworks, the UK government recently decided not to adopt a UK green taxonomy, citing concerns over the resource-intensive nature of taxonomy development and potential interoperability challenges

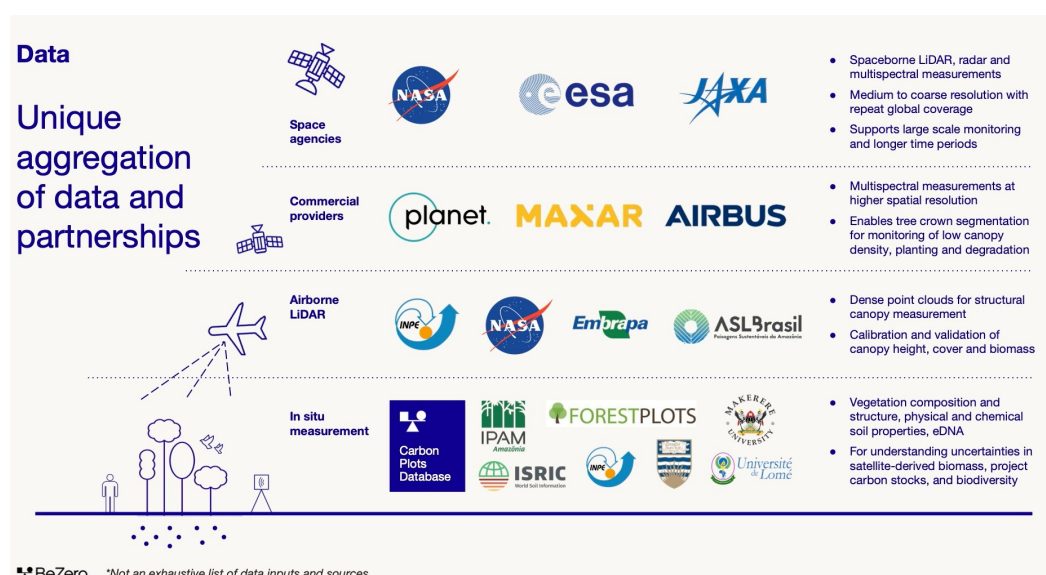


Figure 4: Vertical distribution of BeZero’s aggregated data input from remote sensing to in-situ measurements [50]

with international frameworks [51]. This decision sparked debate among market participants about the role of standardized taxonomies in carbon market development.

Simon argued that this represents a missed opportunity, drawing parallels to the over-the-counter derivatives market where standardization proved crucial for scaling. He noted that the International Swaps and Derivatives Association (ISDA) OTC Derivatives Taxonomy was instrumental in transforming the derivatives market into a multi-trillion dollar asset class. Given that carbon markets are similarly dominated by OTC trading, Simon contended that standardized taxonomy development is a prerequisite for establishing carbon credits as a legitimate asset class.

4.3.2 MRV and Disclosure Standardization

All respondents agree that standardized measurement, reporting, and disclosure mechanisms are beneficial to the voluntary carbon market. However, disagreement emerged regarding the extent of standardization in MRV and disclosures, reflecting a fundamental tension between market-driven innovation and centralized coordination.

Joel emphasized the importance of government adoption of existing market innovations. He argued that *“the government needs to adopt market tools, like ratings and insurance, and make places for the market to adopt them.”* Joel distinguished between compliance-driven approaches that focus on cost minimization and voluntary

market incentives that drive innovation: *“you want to create that incentive to go further and better, and invest more in technology, invest more in improved processes.”* In contrast, Jaye advocates for a more fundamental restructuring that addresses what she sees as critical governance gaps in current carbon market arrangements. She highlighted that *“governments not keeping records of the project-based solutions taking place within their legislative borders”* creates problematic data fragmentation and undermines national sovereignty over natural resources. Jaye argued that the current consultant-driven market *“creates data silos from domestic governments, burning dollars before ground implementation,”* resulting in inefficient resource allocation and limited governmental oversight.

Jaye’s proposed resolution centers on a state-centric approach that does not dismiss market influences but focuses technological innovation on GovTech solutions, technology that enables centralized queries of data such that information can be tracked from the country level down to the project and community level. This would enable governments to ensure traceability and transparency within projects on their land, shifting from market-led fragmentation toward sovereign oversight of domestic carbon resources while maintaining space for market-driven efficiency improvements.

Despite their different approaches to implementation, both perspectives converge on the necessity of increased government involvement in carbon market standardization. Joel advocates for government adoption and integration of market-developed tools and mechanisms, while Jaye proposes government-led technological infrastructure that maintains sovereign control over domestic carbon resources.

4.3.3 Contract Standardization

Another standardization strategy within the VCM involves standardized contracts, which aim to streamline carbon credit transactions by providing uniform terms and conditions. Climate Impact X, for example, has developed standardized contracts that reduce negotiation time and transaction costs by creating consistent frameworks for carbon credit trading [52].

The rationale behind standardized contracts mirrors that of traditional commodity markets, by creating fungible instruments that can be traded efficiently at scale. Proponents argue that standardization reduces friction, lowers transaction costs, and increases market liquidity by making carbon credits more comparable and tradeable across different projects and jurisdictions.

However, Scott pointed out one potential caveat in this approach. He argued against standardizing into formalized contracts, suggesting the market should *“not standard-*

ize into a formalized contract and recognize that this is much more like corporate credit where each bond is different [...] and therefore each credit will be different, and that differential will be addressed by a price.” Scott emphasized that projects should be priced according to their specific characteristics: *“one project will price better or worse than the one next to it because it has a greater likelihood of delivering a ton of carbon [...] or it’s in a country where you’re more certain of getting your carbon, whereas if it was an emerging country with questionable laws, maybe you’d be less likely, and so you would penalize it on price.”*

This perspective highlights a critical concern with standardized contracts: the potential for a “cheapest-to-deliver” problem. When contracts are standardized, buyers may systematically choose the lowest-quality credits that still meet minimal contractual requirements, potentially undermining market integrity and environmental effectiveness. This runs counter to the market’s need for price differentiation that rewards higher-quality projects and creates incentives for improvement.

These considerations suggest that while standardized contracts may represent a viable option for future market development, their implementation may be premature at the current stage. The market may benefit more from allowing natural price differentiation to emerge before introducing standardization mechanisms that could inadvertently commoditize credits regardless of their underlying quality.

5 Recommendations

5.1 Prioritizing Data Integrity as Foundation

For those looking to innovate within the voluntary carbon market, securing data integrity is paramount before pursuing other technological solutions. As demonstrated in the results, all other technological applications, from blockchain to artificial intelligence, depend fundamentally on high-quality, reliable data inputs. Independent third-party validation plays a crucial role in ensuring credibility, but the consultant-driven model has inadvertently created information silos, as each firm applies its own proprietary methodologies.

Innovators should therefore focus not on replacing validation, but on complementing it with solutions that promote standardization and accessibility. This includes developing systems capable of aggregating and structuring the vast amounts of unstructured documentation currently produced in carbon projects, while ensuring analytical independence from commercial interests. Rather than creating proprietary methods that further entrench silos, technology providers should prioritize open data standards and interoperable systems that can integrate across registries and project types. In this way, validation continues to provide trust, while standardization unlocks transparency and scalability.

5.2 Government Technology (GovTech) Solutions

Entrepreneurial individuals should tap into the significant potential for GovTech innovation in facilitating governments, particularly those in the global south. The root cause of the consultant-driven state of the market stems from governments' inability to respond to international strategy changes and develop policy in a timely manner under insufficient capacity, which has resulted in international companies entering their jurisdiction and effectively taking a piece of the pie.

Technology providers should develop GovTech solutions that enable centralized data queries, allowing information to be tracked from country level down to individual projects and communities. These systems should support governments in maintaining sovereign control over their natural resources while facilitating participation in global carbon markets. Such solutions could address the fundamental governance gaps by enabling timely policy responses to international developments and reducing dependence on external consultants. Avoiding overcomplication of the market and ensures community compensation. This represents a pathway for countries to access billions of dollars in annual carbon finance while maintaining domestic oversight

and control.

5.3 Community-Centered Technology Development

The core concept of a “just transition” highlighted in the Paris Agreement [4] underscores the importance of ensuring that climate action does not compromise workforce well-being or local livelihoods. In the context of carbon projects, technological solutions should be designed to create employment opportunities that enable local communities to earn stable income, while including provisions for skills development.

REDD+ projects illustrate this challenge clearly. When projects restrict deforestation through enforcement of protection boundaries without providing alternative livelihoods, they often generate activity-shifting leakage, where local communities dependent on exporting forest resources simply relocate their activities to unprotected areas [53].

Technology providers should develop systems that enable local communities to earn income through participating in monitoring and verification processes. This includes creating accessible platforms for collecting ground-truth data, training programs for local data collectors, and payment systems that fairly compensate community participation. Such approaches can enhance data quality while ensuring that technological advancement contributes to sustainable development rather than exploitation.

Strategic project selection also matters for community benefits. As discussed in Section 4.2.4, certain project types create substantially more local co-benefits than others. Biochar and waste management projects, for instance, provide immediate agricultural improvements and address local waste challenges, while also creating opportunities for local production and distribution networks. Innovation should prioritize technologies that generate tangible, immediate co-benefits alongside carbon sequestration.

The research findings emphasize that successful technology deployment must consider local contexts and create value for communities hosting carbon projects. Solutions that extract value without providing local benefits risk perpetuating the problematic dynamics already present in consultant-driven markets.

5.4 Other recommendations

Disclosure of credit and allowance ownership in corporate balance sheets is crucial for giving investors sufficient data to evaluate whether a company’s climate commit-

ments are credible. Such disclosure regulation should hopefully materialize following the UK's consultation on sustainability disclosure standards that ended September 17th, but technology providers can also develop alternative means of informing the public about corporate carbon positions.

The convergence of voluntary and compliance carbon markets creates opportunities for technological innovation that serves both market segments. However, innovators should focus on developing systems that support transparent pricing based on project-specific characteristics and quality metrics rather than pursuing premature contract standardization while the market is still developing price differentiation mechanisms.

6 Conclusion

This research examined how technological innovation can address integrity challenges undermining voluntary carbon markets. Through semi-structured interviews with key market participants, the study identified data quality as the fundamental prerequisite for all technological solutions, with consensus that digital monitoring must replace manual verification methods to enable market scaling.

The findings reveal mixed prospects for emerging technologies. While artificial intelligence demonstrates practical applications in data aggregation and regulatory analysis, blockchain adoption faces significant barriers including energy consumption concerns and fragmentation risks. The convergence of voluntary and compliance markets creates new opportunities for government-centered technological solutions, though current market capacity remains limited by insufficient organizational expertise.

The research highlights the critical need for government technology innovations that enable sovereign oversight of domestic carbon resources, moving beyond the consultant-driven model that creates data silos and governance gaps. Community-centered approaches that create local employment while enhancing monitoring capabilities emerge as essential for sustainable technology deployment.

For organizations developing technological solutions, the study recommends prioritizing data integrity infrastructure over blockchain development, focusing on user-friendly platforms that build market expertise alongside participation. Strategic emphasis should target engineered removal solutions and enhanced corporate disclosure systems. Success requires recognizing that technological innovation alone cannot resolve market credibility issues without addressing underlying governance, capacity, and transparency challenges that continue to undermine investor confidence and market growth.

Appendix

Appendix A: Participating Interviewees

Table 2: Summary of semi-structured interviews conducted with carbon market participants

Name	Current Affiliation	Title	Nature of Company	Interview Duration (min)	Words Transcribed
Joel Gould	BeZero Carbon	Markets & Policy Lead	Carbon Ratings Agency	32	5217
Simon Puleston Jones	Emral Carbon	CEO	Brokerage & Advisory	37	5489
Scott Eaton	Carbonplace	CEO	Exchange	34	5418
Jaye Connolly	RippleNami	CEO	Technology Provider	53	8136

Appendix B: Interview Protocol

(Note: The bulleted questions outline thematic areas; in practice, they may be phrased differently.)

Project Overview and Motivation

- Motivation: growing interest in the convergence of technology, entrepreneurship, and climate finance
- Hypothesis: innovative technologies are key to scaling the integrity of carbon credits, provided supportive regulation and governance are in place

Request recording consent + Anonymity preferences

Conceptual Understanding

- What aspects of integrity (e.g., additionality, permanence, transparency, fair pricing) are most important in your field?
- Where do you see the biggest integrity risks today?

Importance of Technological Advancements

- How important do you think technological advancements are for improving the integrity of carbon credits?
- Which factors (regulation, governance, market demand, etc.) most influence whether technology actually improves integrity?

Opportunities for Technology Application

- What technologies have you deployed to improve your workflow?
- What examples of successful technology use have you seen in carbon markets?
- Which technological tools do you think have the greatest potential to enhance carbon credit integrity across different lifecycle stages?

Risks, Challenges, and Limitations

- What are the biggest challenges to adopting these technologies at scale?
- Are costs, governance, or trust the biggest obstacles to technology adoption?
- What technical, financial, or regulatory challenges are slowing adoption?

Future Outlook

- Beyond technology alone, what do you think the state of carbon credit integrity will look like in 5 years?
- What would it take for us to get there?

Conclusion

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